jurimo30 Project Plan

<table>
<tr><th>Working Weeks</th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th><th>7</th><th>8</th><th>9</th><th>10</th><th>11</th><th>12</th><th>13</th><th>14</th><th>15</th><th>16</th><th>17</th><th>18</th><th>19</th><th>20</th><th>21</th><th>22</th><th>23</th><th>24</th><th>25</th><th>26</th></tr>
<tr><td>Start of the week</td><td>19-Nov</td><td>26-Nov</td><td>3-Dec</td><td>10-Dec</td><td>17-Dec</td><td>24-Dec</td><td>31-Dec</td><td>7-Jan</td><td>14-Jan</td><td>21-Jan</td><td>28-Jan</td><td>4-Feb</td><td>11-Feb</td><td>18-Feb</td><td>25-Feb</td><td>4-Mar</td><td>11-Mar</td><td>18-Mar</td><td>25-Mar</td><td>1-Apr</td><td>8-Apr</td><td>15-Apr</td><td>22-Apr</td><td>29-Apr</td><td>6-May</td><td>13-May</td></tr>
<tr><td>Sensing and input working package</td><td>Documentation on the given guides and projects.
Chose main languages and technologies
Create/adapt project plan
Members tasks asignation</td><td colspan="7">Camera handling, preprocessing, noise cancelling, ROIs definition</td><td colspan="9">Define other necessary sensors, define use-case, integration (IMU, distance), preprocessing, noise cancelling.</td><td colspan="8"></td><td>BFMC</td></tr>
<tr><td></td><td></td><td colspan="5"></td><td colspan="11">Define use-case and test given servers information (localisation on map, cars interaction, gps interaction)</td><td colspan="8"></td><td></td></tr>
<tr><td></td><td></td><td colspan="16"></td><td colspan="5">Induce noise on all sensors and systems</td><td colspan="3">Other functionalities and optimizations</td><td></td></tr>
<tr><td>Perception and scene understanding working package</td><td></td><td colspan="3">Lane detection</td><td colspan="5">Intersection detection</td><td colspan="4">Traffic sign detection</td><td colspan="4">Traffic light detection</td><td colspan="8"></td><td></td></tr>
<tr><td></td><td></td><td colspan="10"></td><td colspan="6">Position fusion</td><td colspan="5">Traffic lights detection & classification</td><td colspan="3"></td><td></td></tr>
<tr><td></td><td></td><td colspan="12"></td><td colspan="4">Define objects properties file</td><td colspan="5">Object detection & classification</td><td colspan="3"></td><td></td></tr>
<tr><td></td><td></td><td colspan="12"></td><td colspan="9">Environmental server interaction</td><td colspan="3"></td><td></td></tr>
<tr><td></td><td></td><td colspan="21"></td><td colspan="3">Other functionalities and optimizations</td><td></td></tr>
<tr><td>Behaviour and motion plan working package</td><td></td><td colspan="8">Define project architecture and communication between packages</td><td colspan="4">Define path planning and validation</td><td colspan="4">Define robustness and safety measures</td><td colspan="8"></td><td></td></tr>
<tr><td></td><td></td><td colspan="8"></td><td colspan="8">Define decision making --> priorities of actions and state flow</td><td colspan="8"></td><td></td></tr>
<tr><td></td><td></td><td colspan="16"></td><td colspan="5">Induce noise on systems to valdiate robustness (loss of image, burned image, road search, undefined objects and states)</td><td colspan="3">Other functionalities and optimizations</td><td></td></tr>
<tr><td>Vehicle control working packages</td><td></td><td colspan="8">Lane following and speed control</td><td colspan="4">Intersection navigation</td><td colspan="4">Simple action taking maneuvers (parking, stop for traffic sign, stop for traffic light, stop for pedestrian)</td><td colspan="5">Complex action taking maneuvers (swith lane for static and mobile car, road search)</td><td colspan="3"></td><td></td></tr>
<tr><td></td><td></td><td colspan="4">Develop and test brain <-> simulator communication service (middleware)</td><td colspan="17"></td><td colspan="3">Other functionalities and optimizations</td><td></td></tr>
<tr><td>Final result & Demo</td><td>Team can control the phisical car remotely and the virtual car on the simulator.</td><td colspan="8">Robot can keep a lane, can make a curve</td><td colspan="4">Robot can navigate in intersection</td><td colspan="4">Robot can go on a pre-determined path, stop at stop sign, park at parking sign, slow at crosswalk</td><td colspan="5">While detecting and calculating it's position, the robot can dynamicaly go to specified checkpoint, react to traffic lights, interact with other cars and send environemt data)</td><td colspan="3"></td><td></td></tr>
<tr><td></td><td></td><td colspan="4">Team defines and creates it's own phisical testing environment</td><td colspan="4">Team defines a way of parallel developing and testing</td><td colspan="16"></td><td></td></tr>
<tr><td></td><td></td><td colspan="3">Team installs the virtual testing environment</td><td colspan="18"></td><td colspan="3">Other functionalities and optimizations</td><td></td></tr>
<tr><td>Deadlines</td><td></td><td colspan="2"></td><td>16-Dec</td><td colspan="4"></td><td>20-Jan</td><td colspan="3"></td><td>17-Feb</td><td colspan="3"></td><td>17-Mar</td><td colspan="4"></td><td>21-Apr</td><td colspan="3"></td><td>21-May</td></tr>
<tr><td>Checkpoint</td><td></td><td colspan="2"></td><td>1st report</td><td></td><td>Christmas break</td><td></td><td>Christmas break</td><td>2nd report</td><td colspan="3"></td><td>3rd report</td><td colspan="3"></td><td>Mid-term quality gate</td><td colspan="4"></td><td>4th report</td><td colspan="3"></td><td>Finals</td></tr>
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